AUSTRALIAN MATHS TRUST

2020 AMC

AUSTRALIAN MATHEMATICS COMPETITION

Junior Years 7–8
(Australian school years)

CHINESE VERSION

NAME _____

TIME ALLOWED: 75 MINUTES

INSTRUCTIONS AND INFORMATION

General

1. Do not open the booklet until told to do so by your teacher.
2. NO calculators, maths stencils, mobile phones or other calculating aids are permitted. Scribbling paper, graph paper, ruler and compasses are permitted, but are not essential.
3. Diagrams are NOT drawn to scale. They are intended only as aids.
4. There are 25 multiple-choice questions, each requiring a single answer, and 5 questions that require a whole number answer between 0 and 999. The questions generally get harder as you work through the paper. There is no penalty for an incorrect response.
5. This is a competition not a test; do not expect to answer all questions. You are only competing against your own year in your own country/Australian state so different years doing the same paper are not compared.
6. Read the instructions on the answer sheet carefully. Ensure your name, school name and school year are entered. It is your responsibility to correctly code your answer sheet.
7. When your teacher gives the signal, begin working on the problems.

The answer sheet

1. Use only lead pencil.
2. Record your answers on the reverse of the answer sheet (not on the question paper) by FULLY colouring the circle matching your answer.
3. Your answer sheet will be scanned. The optical scanner will attempt to read all markings even if they are in the wrong places, so please be careful not to doodle or write anything extra on the answer sheet. If you want to change an answer or remove any marks, use a plastic eraser and be sure to remove all marks and smudges.

Integrity of the competition

The AMT reserves the right to re-examine students before deciding whether to grant official status to their score.

Reminder: You may sit this competition once, in one division only, or risk no score.

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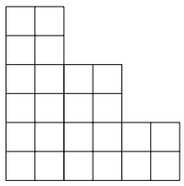
1-10 题，每题 3 分
Questions 1 to 10, 3 marks each

1. 请问右图中总共有多少个 1×1 的正方形？

How many 1×1 squares are in this diagram?



(A) 16 (B) 18 (C) 20 (D) 24 (E) 25

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2. 请问2020 的一半是多少？

What is half of 2020?

(A) 20 (B) 101 (C) 110 (D) 1001 (E) 1010

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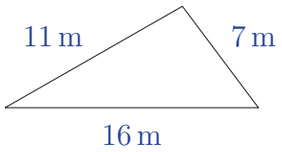
3. 请问右图三角形的周长是多少？

What is the perimeter of this triangle?



(A) 33 m (B) 34 m (C) 35 m (D) 36 m (E) 37 m

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4. 我在上午 8：48 搭上一列火车，并于上午 9：21 下车。请问我在这列火车上待了多少分钟？
I stepped on the train at 8.48 am and got off at 9.21 am. How many minutes did I spend on the train?

(A) 27 (B) 33 (C) 43 (D) 87 (E) 93

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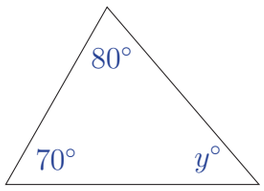
5. 在这个三角形中，请问 y 的值是多少？

What is the value of y in this triangle?



(A) 10 (B) 30 (C) 50 (D) 70 (E) 90

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6. $2 - (0 - (2 - 0)) =$

(A) -4 (B) -2 (C) 0 (D) 2 (E) 4

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7. 如右图，在方格表中，每一行的数字之和标在该行的最右边、每一列的数字之和标在该列的最下边。请问 N 的值是多少？

In the grid, the total of each row is given at the end of the row, and the total of each column is given at the bottom of the column. The value of N is

8		6	16
	N		9
7		9	20
20	7	18	

(A) 1 (B) 2 (C) 3 (D) 4 (E) 5

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8. 将英文字母 **G** 顺时钟旋转 135°。请问下列哪一项可以表示最后呈现的形状？

A letter **G** is rotated clockwise by 135°. Which of the following pictures best represents the final image?

(A) (B) (C) (D) (E)

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9. 计算 $\dfrac{1+2+3+4+5}{1+2+3+4} - \dfrac{1+2}{1+2+3} =$

$\dfrac{1+2+3+4+5}{1+2+3+4} - \dfrac{1+2}{1+2+3} =$

(A) 3 (B) $\frac{5}{6}$ (C) 1 (D) $\frac{7}{6}$ (E) 2

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10. 小莎心里想着两个数，它们的和为 26、差为 14。请问小莎想的这两个数的乘积是多少？

Sebastien is thinking of two numbers whose sum is 26 and whose difference is 14. The product of Sebastien's two numbers is

(A) 80 (B) 96 (C) 105 (D) 120 (E) 132

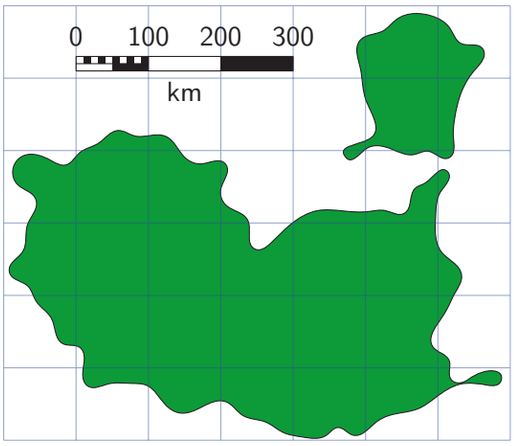
11-20 题，每题 4 分
Questions 11 to 20, 4 marks each

11. 某国的领土包含两座岛屿，如右图所示。请问该国的领土面积为多少平方千米？

(A) 介于 100 与 1000 之间
(B) 介于 1000 与 10 000 之间
(C) 介于 10 000 与 100 000 之间
(D) 介于 100 000 与 1 000 000 之间
(E) 超过 1 000 000

A country consists of two islands as shown on this map.
In square kilometres, its area is

(A) between 100 and 1000
(B) between 1000 and 10 000
(C) between 10 000 and 100 000
(D) between 100 000 and 1 000 000
(E) greater than 1 000 000



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12. $123456 - 12345 + 1234 - 123 + 12 - 1 =$

(A) 33333 (B) 101010 (C) 111111 (D) 122223 (E) 112233

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13. 小莉从出生到现在共经过 2020 天。请问她上次生日是几岁？

Lily is 2020 days old. How old was she on her last birthday?

(A) 4 (B) 5 (C) 6 (D) 7 (E) 8

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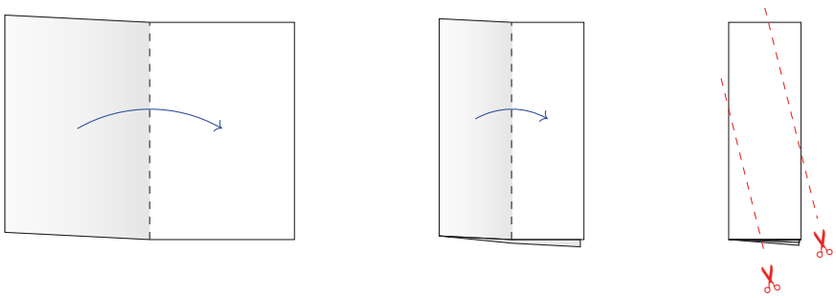
14. 将一张纸片如下图所示折叠两次，然后沿着虚线剪掉两侧部分后，将它展开。



请问最像哪个英文字母？

A piece of paper is folded twice as shown and cut along the dotted lines.

Once unfolded, which letter does the piece of paper most resemble?

(A) M (B) O (C) N (D) B (E) V

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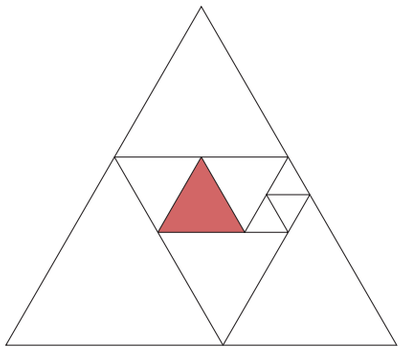
15. 将一个正三角形分割成若干个小正三角形，如图所示。涂上阴影的三角形边长为 2 。请问原来这个大正三角形的周长是多少？

An equilateral triangle is subdivided into a number of smaller equilateral triangles, as shown. The shaded triangle has side length 2. What is the perimeter of the large triangle?



(A) 24 (B) 27 (C) 30

(D) 33 (E) 36

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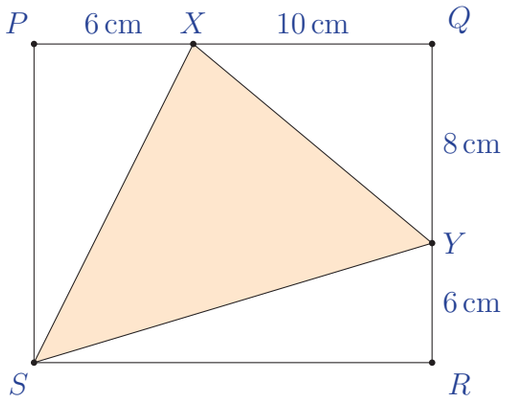
16. 如右图所示，
三角形 XYS 被矩形 PQRS 封闭在内，
请问三角形 XYS 的面积为多少 cm^2？

Triangle XYS is enclosed by rectangle $PQRS$ as shown in the diagram.
In square centimetres, what is the area of triangle XYS?



(A) 82 (B) 88 (C) 94

(D) 112 (E) 130

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17. 四支队伍进行足球循环赛。每支队伍都能和其它三支队伍比赛一次。
胜队可得 3 分、平手各得 1 分、败队得 0 分。当所有比赛结束后，
有一支队伍获得 6 分、有两支队伍各得 4 分、有一支队伍获得 3 分。
请问总共有多少场比赛双方打成平手？
Four teams play in a soccer tournament. Each team plays one game against each of the other three teams. Teams earn 3 points for a win, 1 point for a draw and 0 points for a loss. After all the games have been played, one team has 6 points, two teams have 4 points and one team has 3 points. How many games ended in a draw?

(A) 0 (B) 1 (C) 2 (D) 3 (E) 4

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18. 一个等腰三角形的周长为 28 cm 且各边长均为整数。请问满足条件的三角形有多少种？

An isosceles triangle has a perimeter of 28 cm and sides of integer length. How many different such triangles can be made?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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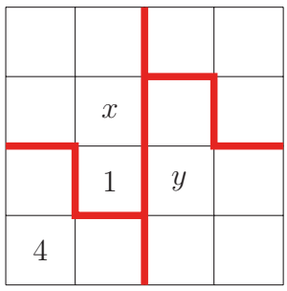
19. 在如图所示的网格中，数字1到4会在每行和每列中出现一次。同时，红线分隔的四个区域中每个区域的数字总和必须相等。
请问 $x+y$ 的和为多少？

In the grid shown, the numbers from 1 to 4 must appear once in each row and column. Also, the sum of numbers in each of the four regions separated by red lines must be the same.
What is the sum $x+y$?



(A) 8 (B) 7 (C) 6 (D) 5 (E) 4

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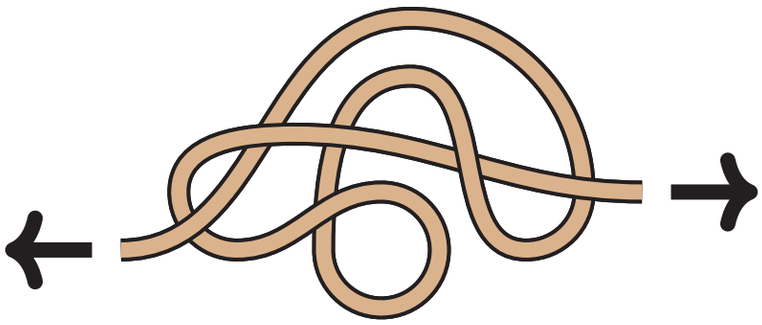
20. 小安有一个周长为 400 cm 的正方形硬纸板。他在硬纸板上画了一条水平线和一条垂直线，然后沿着这两条线将它切分成四个矩形。请问这四个矩形周长之和的最大可能值为多少cm？

Anupam has a cardboard square with a perimeter of 400 centimetres. He draws a horizontal line and a vertical line on the square and cuts along these lines to create four rectangles. What is the largest possible sum of the perimeters of these four rectangles, in centimetres?

(A) 400 (B) 600 (C) 800 (D) 1000 (E) 1200

21-25 题，每题 5 分
Questions 21 to 25, 5 marks each

21. 如图所示，向箭头方向拉动这条缠绕的绳子两端，可以解开绳结或者变成小绳结。



请问下列哪一项最符合这条绳子拉紧后形成的样子？

The ends of the tangled string shown are pulled in the direction of the arrows so that the string either untangles or forms a simpler knot.

Which of the following best matches the knot, or otherwise, that is formed when the string is tightened?

(A) (B) (C)

(D) (E)

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22. 丁老师给学生布置了一些家庭作业来练习数学运算顺序。其中一道题目是 $2+3\times(4+3)$，答案是 23。然而，其中一位学生只是从左到右运算且忽略括号，写下 $2+3=5$，$5\times4=20$，$20+3=23$，答案竟然正确。
丁老师认为这太神奇了，于是他试图造出另一个问题，使得从左到右运算也可以得到正确答案。他尝试 $5+4\times(7+\square)$。请问在小方格内应该填入的数是什么？

Mr Atkins wrote some homework questions for his class to practise order of operations. One of the questions was $2+3\times(4+3)$, with answer 23. However, one of his students just worked from left to right and ignored the brackets, writing $2+3=5$, $5\times4=20$, $20+3=23$, the correct answer.
Mr Atkins thought that this was fascinating, so he tried to come up with another question where working left to right gave the right answer. He tried $5+4\times(7+\square)$. What number should he put in the box?

(A) 2 (B) 4 (C) 6 (D) 8 (E) 10

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23. 我和朋友参加一项有 10 道题目的数学测验。第 1 题占 1 分、第 2 题占 2 分、第 3 题占 3 分，依此类推。答案正确得到该题全部的分数，答案错误该题得 0 分。最后我们两人的得分相同且答对的题目数量也相同，但是，我们答对的题目不完全相同。请问此次测验中，我最多可以得到多少分？

My friend and I took a maths test with 10 questions. Question 1 was worth 1 mark, question 2 was worth 2 marks, question 3 was worth 3 marks, and so on. Correct answers scored full marks and incorrect answers scored 0 marks.
We both scored the same number of marks and correctly answered the same number of questions. However, we didn't solve exactly the same set of questions as each other.
What is the maximum score that I could have received for the test?

(A) 44 (B) 46 (C) 48 (D) 50 (E) 52

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24. 有一座轻轨路网上共有 21 位司机，但在同一时段内他们不需要全部都工作。

- 周五晚上需要 15 位司机轮班。
- 周六早上需要 12 位司机轮班。
- 周日早上需要 9 位司机轮班。

假如每位司机至少要轮一次班，那么最多有多少位司机可以轮三次班？

A light rail network has 21 drivers, but not all of them are required at the same time:

- 15 drivers are required for the Friday night shift.
- 12 drivers are required for the Saturday morning shift.
- 9 drivers are required for the Sunday morning shift.

Given that every driver must work on at least one of these shifts, what is the maximum number of drivers that can work on all three shifts?

(A) 5 (B) 6 (C) 7 (D) 8 (E) 9

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25. 一个袋子内恰好有 50 枚硬币。这些硬币的面额分别为 10 元、20 元或 50 元，且每种面额袋子中至少有一枚。已知袋子中的硬币总金额为 1000 元。
请问袋子中总共有多少种不同情况？

A bag contains exactly 50 coins. The coins are either worth 10 cents, 20 cents or 50 cents, and there is at least one of each. The total value of the coins is $10.
How many different ways can this occur?

(A) 2 (B) 4 (C) 8 (D) 12 (E) 16

问题 26-30 的答案为 000-999 之间的整数，
请将答案填在答案卡上对应的位置。

第 26 题占 6 分，第 27 题占 7 分，第 28 题占 8 分，
第 29 题占 9 分，第 30 题占 10 分。

For questions 26 to 30, shade the answer as an integer from 0 to 999 in the space provided on the answer sheet.

Questions 26–30 are worth 6, 7, 8, 9 and 10 marks, respectively.

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26. 不重复使用数字 1 到 9，恰好可以组成三个 3 位数。已知第二个数是第一个数的三倍、第三个数是第一个数的五倍，请问第二个数是多少？

The digits 1 to 9 are used exactly once each to make three 3-digit numbers. The second number is three times the first number. The third number is five times the first number. What is the second number?

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27. 小美在这个键盘输入她的三位验证码 (PIN)。这三个数字各不相同，但第一个数与第二个数的按键相邻、第二个数与第三个数的按键也相邻。
例如，563 是一个可能的验证码，但是 536 则不是，因为按键 5 与 3 不相邻。
请问小美总共有多少个可能的验证码？

Madeleine types her three-digit Personal Identification Number (PIN) into this keypad.
All three digits are different, but the buttons for the first and second digits share an edge, and the buttons for the second and third digits share an edge.
For instance, 563 is a possible PIN, but 536 is not, since 5 and 3 do not share an edge.
How many possibilities are there for Madeleine's PIN?

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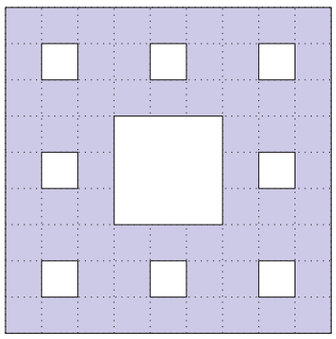
28. 从一个 $9 \times 9 \times 9$的正立方体开始。小欧从中移除尽可能少的 $1 \times 1 \times 1$正立方体，使得最后立方体的正视图、俯视图和左视图都与右图所示相同。请问小欧总共移除了多少个 $1 \times 1 \times 1$ 的正立方体？



Starting with a $9 \times 9 \times 9$ cube, Augusta removed as few $1 \times 1 \times 1$ cubes as possible so that the resulting sculpture had front view, top view and side view all the same, as shown. How many $1 \times 1 \times 1$ cubes did Augusta remove?

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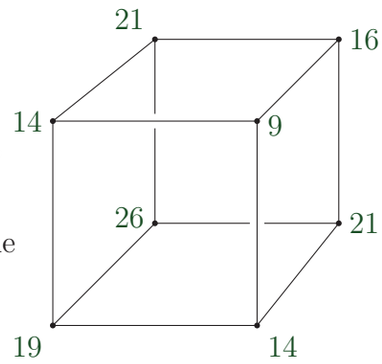
29. 在一个正立方体的每个表面都放置 1 到 10 之中的一个整数，它们互不相同，然后在每个顶点上标记交会在此顶点的三个平面上的数之和。
右图只显示了各顶点上的数。
请问在各表面上的数之中，最小的四个数的乘积是多少？

A different integer from 1 to 10 is placed on each of the faces of a cube. Each vertex is then assigned a number which is the sum of the numbers on the three faces which touch that vertex.
Only the vertex numbers are shown here.
What is the product of the 4 smallest face numbers?



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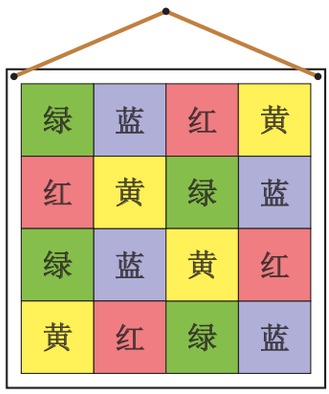
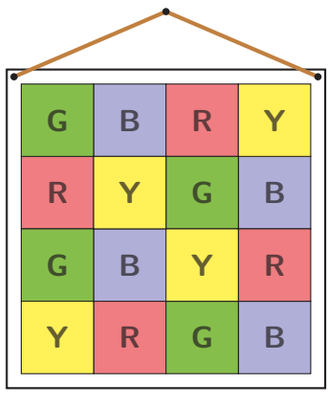
30. 我将 16 片正方形碎布缝合在一起制作一个 4×4 的方格表挂饰，碎布颜色分为红色、蓝色、绿色、黄色四种，要求同颜色的碎布不可以接触，包括顶点。

右图是一个失败的试验品，因为两片黄色碎布的顶点有接触。

请问我总共有多少种不同的方法可以正确排列这些碎布？



My grandson makes wall hangings by stitching together 16 square patches of fabric into a 4×4 grid. I asked him to use patches of red, blue, green and yellow, but to ensure that no patch touches another of the same colour, not even diagonally.

The picture shows an attempt which fails only because two yellow patches touch diagonally.

In how many different ways can my grandson choose to arrange the coloured patches correctly?